

Multiple Choice (10 marks)

1. Scientists who disagree with the results of an experiment should
 - (a) change the experiment.
 - (b) keep their opinions to themselves.
 - (c) find out what other scientists think about the results.
 - (d) repeat the experiment several times and compare results.
2. Large blooms of algae on the surface of a lake keep which abiotic factor from reaching the bottom?
 - (a) water
 - (b) salt
 - (c) light
 - (d) oxygen
3. Which tool would be best to use to determine how long it takes a cup of water to boil?
 - (a) balance
 - (b) hot plate
 - (c) stopwatch
 - (d) thermometer
4. Adaptive characteristics that help dolphins survive life in the ocean include all of these except
 - (a) their sleek skin.
 - (b) swimming very fast.
 - (c) their body shape.
 - (d) traveling alone.
5. What information best helps a meteorologist predict the possibility of tornadoes developing in a certain area?
 - (a) current wind conditions
 - (b) type of clouds in the sky
 - (c) percent humidity in the air
 - (d) amount of rainfall

True / False (10 marks)

Answer True or False

6. True or False: Succession is the process that establishes a community on exposed bedrock after a glacier melts and retreats.
7. True or False: The height of the players should be kept constant to accurately assess their throwing abilities.
8. True or False: The substance started off as a solid, which expanded before melting when heat was applied.
9. True or False: Evaporation occurs at the highest rate in estuaries due to their warm temperatures and high surface area.
10. True or False: Electricity can be used to heat an oven, making it useful for cooking food.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the geological processes that lead to the formation of fossil fuels like coal and natural gas, and analyze their implications for energy production and the environment.
12. After the farmer gathers the crops, analyze the impact this has on the habitat of mice in the field and discuss how it may affect the owl's hunting behavior.

End of Paper